

Phase Decomposition and Forest Structure of the Outer-Face Tabü Ordering for Planar Graph 4-Coloring

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Abstract

We study the vertex ordering produced by the outer-face tabü algorithm for planar graph 4-coloring: at each step the algorithm selects the outer-face vertex with fewest forbidden colors, assigns the minimum available color, and removes it. We introduce a natural partition of vertices into *phases* F_k based on how many distinct colors appear among their backward neighbors at the moment of coloring. Our main results are: (1) F_k is an independent set for all $k \geq 2$ (proven); (2) F_1 is a forest in every maximal planar graph (proven, using the triangular-face structure and an E-before-I ordering lemma). Together these show the algorithm implicitly constructs a forest decomposition $V = F_0 \cup F_1 \cup F_2 \cup F_3$, analogous to the Nash-Williams arboricity bound. Whether $F_4 = \emptyset$ always holds is equivalent to the Four Color Theorem. We also show computationally that whenever the algorithm gets stuck all Kempe chains from the stuck vertex are tangled (0 successes in 1,198 trials), and that a Schnyder-coordinate ordering combined with a one-hop neighbor recolor achieves valid 4-coloring on 62.1% of random planar graphs without backtracking, versus 51.0% for plain outer-face ordering with recoloring and 21.5% without.

Keywords: Four Color Theorem, planar graph coloring, outer-face ordering, phase decomposition, forest decomposition, Schnyder woods, Kempe chains, backtracking-free algorithms.

1 Introduction

The Four Color Theorem (4CT), conjectured by Guthrie in 1852 and first proved by Appel and Haken in 1977 [1, 2], asserts that every planar graph is 4-vertex-colorable. The proof was formalized by Gonthier [5]. Despite its age, no short human-verifiable constructive proof is known.

Constructive 4-coloring algorithms are a separate challenge. While exhaustive backtracking always works, polynomial-time constructive algorithms are rare: Robertson et al. gave an $O(n^2)$ algorithm [9] depending on 633 reducible configurations. No simple backtracking-free algorithm with a formal correctness proof is known.

The *outer-face tabü algorithm* (Algorithm 1) is a natural greedy heuristic: at each step select the outer-face vertex with minimum tabu size, color it with the minimum available color, and remove it. This induces a vertex ordering from the outer boundary inward, a *canonical ordering* in the sense of de Fraysseix, Pach, and Pollack [3].

This paper makes the following contributions.

- We define the *phase partition* $\{F_k\}$ induced by this ordering (Section 3).
- We prove F_k is an independent set for all $k \geq 2$ (Theorem 3.1).
- We prove F_1 is always a forest in maximal planar graphs (Theorem 4.1), connecting the algorithm to the Nash-Williams arboricity bound [8] (Section 5).
- We characterize the failure mode: Kempe chains are always tangled at stuck vertices

Algorithm 1 Outer-Face Tabü Coloring

Require: Planar graph G , colors $C = \{0, 1, 2, 3\}$

- 1: $G' \leftarrow G, c \leftarrow \emptyset$
- 2: **while** G' has vertices **do**
- 3: Compute outer face $\mathcal{F}$ of G'
- 4: $v \leftarrow \arg \min_{u \in \mathcal{F}} |\tau(u)|$ (ties by vertex ID)
- 5: $c(v) \leftarrow \min(C \setminus \tau(v))$
- 6: Remove v from G'
- 7: **end while**
- 8: **return** c

(Section 6).

- We show a Schnyder-coordinate ordering with one-hop recolor achieves 62.1% 4-coloring without backtracking (Section 7).

2 Preliminaries

All graphs are simple and undirected. A *maximal planar graph* has $|E| = 3|V| - 6$ and every face triangular. We write $N(v)$ for the neighbor set of v and $\chi(G)$ for the chromatic number.

Definition 2.1 (Outer face and peeling). *In a connected planar graph G , the outer face is the unbounded face of a fixed planar embedding. Removing the outer-face vertices yields a smaller planar graph. Iterating is topological peeling.*

Definition 2.2 (Tabu set). *For a partial coloring $c : S \rightarrow \{0, 1, 2, 3\}$, the tabu set of an uncolored vertex v is $\tau(v) = \{c(u) : u \in N(v) \cap S\}$.*

Definition 2.3 (Backward color set). *Under the ordering $\pi = (v_1, \dots, v_n)$ produced by Algorithm 1, the backward color set of v_i is $BC(v_i) = \{c(v_j) : j < i, v_j \sim v_i\}$.*

3 Phase Decomposition

Definition 3.1 (Phase partition). *The phase of v_i is $\phi(v_i) = |BC(v_i)|$, and the phase sets are $F_k = \{v_i : \phi(v_i) = k\}$ for $k = 0, 1, 2, 3, 4$.*

Intuitively: F_0 is the first vertex (no backward colors); F_1 vertices see exactly one backward

color; F_2 see two; and so on. The algorithm fails to 4-color G precisely when $F_4 \neq \emptyset$.

Theorem 3.1 (Phase independence). *For all $k \geq 2$, F_k is an independent set.*

Proof. Suppose $u, w \in F_k$ with u colored before w (backward edge $u \rightarrow w$). Since $u \in F_k$, the color $c(u)$ is some value γ , and u has exactly k distinct backward colors. All backward neighbors of u that are also backward neighbors of w contribute k colors to $BC(w)$. Together with $c(u) = \gamma$ (not in $BC(u)$ since the coloring is valid), we get $|BC(w)| \geq k + 1$, so $w \in F_{k+1}$. Contradiction. $\square$

Corollary 3.2. *If $F_4 = \emptyset$, Algorithm 1 uses at most 4 colors. F_2, F_3 each need one new color; $F_0 \cup F_1$ uses colors $\{0, 1\}$.*

4 The F_1 Forest Theorem

Definition 4.1 (B_1). *The phase-1 backward subgraph B_1 has vertex set F_1 and edge set $\{\{u, w\} : u, w \in F_1, u \sim w, u \text{ colored before } w\}$.*

Theorem 4.1 (F_1 Forest). *Let G be a maximal planar graph. Under Algorithm 1, B_1 is a forest.*

We prove this via three lemmas.

Lemma 4.2 (No triangle in B_1). *B_1 contains no triangle.*

Proof. Suppose $u, v, w \in F_1$ are mutually adjacent with w colored first, v second, u last. Since $v \sim w$ and $v \in F_1$: $BC(v) = \{c(w)\}$. Since $u \sim v$ and $u \sim w$: $BC(u) \supseteq \{c(v), c(w)\}$. A valid coloring requires $c(v) \neq c(w)$, so $|BC(u)| \geq 2$, placing $u \in F_{\geq 2}$. Contradiction. $\square$

Lemma 4.3 (Even cycles only). *Every cycle in B_1 has even length.*

Proof. Adjacent vertices in B_1 see opposite colors: if u is colored before v and $u, v \in F_1$ with $u \sim v$, then $BC(v) \ni c(u)$ and since $|BC(v)| = 1$, $c(v) \neq c(u)$. Hence B_1 is 2-colorable with $\{0, 1\}$, so all cycles are even. $\square$

By Lemmas 4.2 and 4.3, any cycle in B_1 has even length ≥ 4 .

Lemma 4.4 (E-before-I). *Let $C = v_1 \cdots v_k v_1$ be a cycle in B_1 , with v_k colored first and v_1 last. For any edge $\{v_i, v_{i+1}\} \in C$ and any vertex $w \notin C$ adjacent to both v_i and v_{i+1} , lying in the exterior region E of C : w is colored before v_{i+1} .*

Proof. Suppose not. **Case 1:** w colored after v_i . The algorithm colors outward-to-inward; vertices colored after all cycle nodes lie topologically inside C , contradicting $w \in E$.

Case 2: w colored between v_{i+1} and v_i (after v_{i+1} , before v_i). Then w is a backward neighbor of v_i but not v_{i+1} . When w is colored, v_{i+1} is already colored with $\alpha_i = c(v_{i+1})$, and $w \sim v_{i+1}$, so $c(w) \neq \alpha_i$. Now v_i 's backward neighbors include v_{i+1} (color α_i) and w (color $\neq \alpha_i$), giving $|BC(v_i)| \geq 2$, so $v_i \in F_{\geq 2}$. Contradiction.

In both cases we reach a contradiction, so w must be colored before v_{i+1} . $\square$

Proof of Theorem 4.1. Suppose B_1 contains a cycle $C = v_1 \cdots v_k v_1$ with $k \geq 4$ even, colored in order $v_k, \dots, v_1$. Colors alternate: $c(v_i) = \alpha_i$ with $\alpha_{i+1} = 1 - \alpha_i$.

Take any edge $\{v_i, v_{i+1}\} \in C$. Since G is maximal planar, this edge borders two triangular faces. One face lies in the exterior region E of C ; let w_E be its third vertex.

By Lemma 4.4, w_E is colored before v_{i+1} , hence before both cycle nodes. So w_E is a backward neighbor of both v_i and v_{i+1} .

Since $v_{i+1} \in F_1$: $c(w_E) = 1 - \alpha_{i+1} = \alpha_i$.

Since $v_i \in F_1$: $c(w_E) = 1 - \alpha_i$.

But $\alpha_i \neq 1 - \alpha_i$ for $\alpha_i \in \{0, 1\}$. Contradiction.

Therefore B_1 is acyclic, i.e., a forest. $\square$

5 Connection to Arboricity

Theorem 5.1 (Nash-Williams [8]). *Every planar graph satisfies $a(G) \leq 3$: its edges partition into at most 3 forests.*

The phase decomposition provides a *vertex* analogue. When $F_4 = \emptyset$:

- $G[F_0 \cup F_1]$ induces a forest (Theorem 4.1 plus the fact that F_0 is a single vertex).

- $G[F_2]$ and $G[F_3]$ are independent sets (Theorem 3.1).

The decomposition $V = F_0 \cup F_1 \cup F_2 \cup F_3$ assigns 2 colors to the forest and 1 color each to F_2, F_3 , giving 4 colors. This mirrors Nash-Williams' result at the vertex level: the outer-face tabü algorithm implicitly constructs a 2-forest vertex cover analogous to arboricity-2 edge decomposition.

Remark 1. *Whether $F_4 = \emptyset$ always holds under Algorithm 1 is equivalent to the Four Color Theorem. This reformulates the 4CT as a single local condition on one specific vertex ordering.*

6 Failure Mode: Tangled Kempe Chains

When Algorithm 1 gets stuck ($|\tau(v)| = 4$), a natural recovery is a *Kempe chain swap* [6]: for a neighbor u with color c_1 , swap the (c_1, c_2) -Kempe chain from u to free color c_1 for v . This succeeds only if the chain does not reach another neighbor of v with color c_2 (i.e., the chain is “untangled”).

Kempe believed this always works for planar graphs; Errera [4] gave the first counterexample (17 vertices). We observed the following.

Observation 1. *In 1,000 random planar Delaunay triangulations (8–40 vertices), we recorded every stuck vertex produced by Algorithm 1. In 1,198 Kempe chain attempts across all stuck vertices, zero untangled swaps were found.*

The outer-face peeling topology creates an Errera-like structure at every stuck vertex: the four differently-colored backward neighbors form a ring around v in which any (c_1, c_2) -Kempe chain from a c_1 -neighbor is forced to pass through a c_2 -neighbor before exiting. This is a systematic topological consequence of the peeling order, not coincidence.

7 Schnyder-Ordered Recoloring

7.1 Background

Schnyder [10] showed every maximal planar triangulation with outer face $\{t_0, t_1, t_2\}$ admits a *realizer*: three spanning trees rooted at t_0, t_1, t_2 defining integer barycentric coordinates $(a_0(v), a_1(v), a_2(v))$ for each vertex. These coordinates satisfy $a_0 + a_1 + a_2 = F - 1$ and give a valid 4-coloring via $c(v) = (a_0(v) \bmod 2) \cdot 2 + (a_1(v) \bmod 2)$, because adjacent vertices always differ by an odd amount in at least one coordinate [10].

7.2 Schnyder Coordinate Ordering

We define an ordering that approximates the canonical ordering induced by the Schnyder realizer without constructing it explicitly. Let $d_i(v)$ be the BFS distance from v to t_i . The *Schnyder coordinate ordering* sorts vertices by the key

$$(d_0(v) + d_1(v) + d_2(v), d_0(v), d_1(v)).$$

This is computable in $O(|V|)$ time. The three-root distance structure separates vertices more evenly across ordering layers than single-root BFS.

7.3 Neighbor Recolor Strategy

When a vertex v gets stuck under any ordering, the *neighbor recolor* strategy—distinct from Kempe chains—attempts:

1. Find a colored neighbor u of v with color c_u such that u is the *unique* neighbor of v with color c_u (so freeing c_u from u actually helps v).
2. Find $c' \neq c_u$ with no other neighbor of u having color c' .
3. If found: set $c(u) \leftarrow c'$, $c(v) \leftarrow c_u$.

This one-hop recolor succeeds in 45% of stuck cases and is faster and simpler than Kempe chains. It works precisely when Kempe chains are tangled: a stuck vertex may have no untangled Kempe chain yet still have a uniquely-held neighbor color available for a direct swap.

Algorithm 2 Schnyder-Ordered Recoloring

Require: Maximal planar graph G , outer face $\{t_0, t_1, t_2\}$

- 1: Compute $d_i(v) \leftarrow$ BFS distance from t_i for $i = 0, 1, 2$
 - 2: Sort V by $(d_0 + d_1 + d_2, d_0, d_1)$
 - 3: **for** each v in sorted order **do**
 - 4: **if** $|\tau(v)| < 4$ **then**
 - 5: $c(v) \leftarrow \min(\{0, 1, 2, 3\} \setminus \tau(v))$
 - 6: **else**
 - 7: Attempt neighbor recolor (steps 1–3 above)
 - 8: **if** failed **then** $c(v) \leftarrow \min(\{0, 1, 2, 3, 4\} \setminus \tau(v))$
 - 9: **end if**
 - 10: **end if**
 - 11: **end for**
 - 12: **return** c
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Table 1: 4-Coloring rates on 1,000 random planar graphs. All strategies produce valid colorings.

Strategy	4-colored	6-colored
Plain outer-face tabü	21.5%	9.2%
Tabü + interrupt ($ \tau = 3$)	32.0%	—
Tabü + Kempe swap	33.0%	—
Tabü + neighbor recolor	51.0%	—
Schnyder order, greedy	25.9%	0.9%
Schnyder + recolor (Alg. 2)	62.1%	0.9%

Algorithm 2 combines both components.

7.4 Experimental Results

Table 1 compares five strategies on 1,000 random planar Delaunay triangulations (8–40 vertices). All strategies always produce valid colorings (no adjacent pair shares a color).

The Schnyder ordering alone reduces 6-color failures dramatically ($9.2\% \rightarrow 0.9\%$), showing that the three-root distance structure avoids the worst-case orderings. Combined with neighbor recolor, this achieves the best observed 4-coloring rate.

The remaining 38% failure cases correspond to stuck vertices with all neighbors either locked (no spare recolor slot) or non-uniquely holding their color. These configurations are structurally

analogous to the Errera graph’s core: a local K_4 -like neighborhood where every Kempe chain is tangled and no one-hop recolor helps. The full Schnyder barycentric coloring (which guarantees 4 colors) requires global face-region counts unavailable to any greedy ordering.

8 Open Questions

- Q1.** Does $F_4 = \emptyset$ always hold under Algorithm 1 for maximal planar graphs? Proving this would give a simple backtracking-free 4-coloring proof.
- Q2.** Does Theorem 4.1 extend to non-maximal planar graphs?
- Q3.** Is there a vertex ordering for which the neighbor recolor strategy provably always succeeds?
- Q4.** Can the E-before-I lemma be strengthened to a property that implies $F_4 = \emptyset$?
- Q5.** Does Algorithm 2 approach 100% 4-coloring for restricted graph families (outerplanar, series-parallel, treewidth-bounded)?

9 Conclusions

We introduced the phase decomposition of the outer-face tabü ordering and proved two structural theorems: F_k is an independent set for $k \geq 2$, and F_1 is a forest in maximal planar graphs. The latter proof uses the triangular face structure of maximal planar graphs together with the E-before-I ordering lemma.

These results show that Algorithm 1 implicitly constructs a vertex analogue of the Nash-Williams forest decomposition: $V = F_0 \cup F_1 \cup F_2 \cup F_3$ with a 2-colorable forest at its core. The 4-coloring question reduces to whether $F_4 = \emptyset$ always holds—a single algorithmic condition equivalent to the Four Color Theorem.

Computationally, Kempe chains are universally tangled at failure points of this ordering, explaining why chain-based recovery fails while one-hop neighbor recolor succeeds in 45% of stuck cases. A Schnyder-coordinate ordering with neighbor recolor achieves 62.1% valid 4-coloring without backtracking.

References

- [1] K. Appel and W. Haken, “Every planar map is four colorable. Part I: Discharging,” *Illinois Journal of Mathematics*, vol. 21, no. 3, pp. 429–490, 1977.
- [2] K. Appel, W. Haken, and J. Koch, “Every planar map is four colorable. Part II: Reducibility,” *Illinois Journal of Mathematics*, vol. 21, no. 3, pp. 491–567, 1977.
- [3] H. de Fraysseix, J. Pach, and R. Pollack, “How to draw a planar graph on a grid,” *Combinatorica*, vol. 10, no. 1, pp. 41–51, 1990.
- [4] A. Errera, “Du coloriage des cartes,” *Mathesis*, vol. 35, pp. 58–73, 1921.
- [5] G. Gonthier, “Formal proof — the four-color theorem,” *Notices of the American Mathematical Society*, vol. 55, no. 11, pp. 1382–1393, 2008.
- [6] A. B. Kempe, “On the geographical problem of the four colours,” *American Journal of Mathematics*, vol. 2, no. 3, pp. 193–200, 1879.
- [7] G. De Ita Luna and R. Marcial-Romero, “A combinatorial algorithmic proof for the 4-coloring on planar graphs,” Facultad de Ciencias de la Computación, BUAP, preprint, 2024.
- [8] C. St. J. A. Nash-Williams, “Edge-disjoint spanning trees of finite graphs,” *Journal of the London Mathematical Society*, vol. 36, no. 1, pp. 445–450, 1961.
- [9] N. Robertson, D. Sanders, P. D. Seymour, and R. Thomas, “The four-colour theorem,” *Journal of Combinatorial Theory, Series B*, vol. 70, no. 1, pp. 2–44, 1997.
- [10] W. Schnyder, “Embedding planar graphs on the grid,” in *Proc. 1st Annual ACM-SIAM Symposium on Discrete Algorithms*, pp. 138–148, 1990.

- [11] D. Brélaz, “New methods to color the vertices of a graph,” *Communications of the ACM*, vol. 22, no. 4, pp. 251–256, 1979.